

CONIFERS

P. peuce, *P. ponderosa*, *Sciadopitys*, *Sequoia*, *Sequoiadendron* and *Taxodium* will grow on heavy clay soils. *Picea omorika*, *P. sitchensis*, *Pinus contorta*, *Sciadopitys verticillata* and *Thuja plicata* are all happy on wet soil, and *Metasequoia* and *Taxodium* thrive in waterlogged conditions. *Cupressus*, *Juniperus* and *Pinus* grow well on dry, sandy soil.

Pruning

If a conifer produces more than one leader, remove all but one. Bear in mind when trimming hedges that most conifers will not produce new growth when cut back into old wood or from branches that have turned brown. This does not, however, apply to *Cephalotaxus*, *Cryptomeria*, *Cunninghamia*, *Sequoia*, *Taxus*, *Torreya*, and *Wollemia*; these conifers may be kept to a reasonable size by cutting back the main stem, which will later coppice (produce new growth). Young specimens of *Araucaria*, *Ginkgo*, *Metasequoia* and *Taxodium* will sometimes do the same.

Propagation

Seed is the easiest method of propagation, but forms selected for leaf color (other than blue in some species) do not come true. Sow in fall or spring. All genera apart from *Abies*, *Cedrus*, *Picea* (except young plants or dwarf forms), *Pinus*, *Pseudolarix*, *Pseudotsuga* and *Tsuga* (except young plants or dwarf forms) may be raised fairly easily from cuttings: current growth from fall to spring for evergreens, softwood cuttings in summer for deciduous conifers. Tall-growing forms of Pinaceae (*Abies*, *Cedrus*, *Picea*, *Pinus*, *Pseudolarix*, *Pseudotsuga* and *Tsuga*) are usually propagated by grafting in late summer, winter or early spring. Layering may be possible for some dwarf conifers. It is illegal to propagate *Wollemia nobilis*.

Pests and diseases

Honey fungus attacks many conifers, especially young plants. Most resistant to the disease are *Abies*, *Calocedrus*, *Larix*, *Pseudotsuga* and *Taxus*. Green spruce aphid may be a problem on *Picea*, and conifer spinning mite may defoliate *Abies*, *Picea* and some *Pinus*.

Conifers are illustrated on pp.95–105, dwarf forms on pp.105. See also *Abies*, *Araucaria*, *Athrotaxis*, *Austrocedrus*, *Calocedrus*, *Cedrus*, *Cephalotaxus*, *Chamaecyparis*, *Cryptomeria*, *Cunninghamia*, *Cupressus*, x *Cuprocyparis*, *Fitzroya*, *Ginkgo*, *Juniperus*, *Larix*, *Metasequoia*, *Microbiota*, *Phyllocladus*, *Picea*, *Pinus*, *Podocarpus*, *Pseudolarix*, *Pseudotsuga*, *Saxegothaea*, *Sciadopitys*, *Sequoia*, *Sequoiadendron*, *Taxodium*, *Taxus*, *Thuja*, *Thujopsis*, *Torreya*, *Tsuga*, and *Wollemia*.

CONOPHYTUM

SYN. OPHTHALMOPHYLLUM

AIZOACEAE

Genus of dwarf, often slow-growing, mat- or clump-forming perennial succulents with spherical or 2-eared leaves that grow for only 2 months each year, after flowering. In early summer, old leaves gradually shrivel to papery sheaths from which new leaves and flowers emerge in

late summer. Frost tender if dry. Needs full sun and well-drained soil. Keep dry in winter. Propagate by seed from spring to fall or by division in late summer. **C. bilobum** illus. p.494. **C. concordans**, syn. *Ophthalmophyllum villetii*, illus. p.484. **C. frutescens**, syn. *C. notabile*, illus. p.496. **C. longum**, syn. *Ophthalmophyllum herri*, *O. longum*. Clump-forming, perennial succulent. H 1¼in (3cm), S ½in (1.5cm). Has 2 almost united, cylindrical, very fleshy, erect, gray-green to brown leaves. Bears daisylike, white to pink flowers, ¾in (2cm) across, in late summer. Z10–11 (min. 41°F/5°C) **C. notabile**. See *C. frutescens*. **C. truncatum**. Slow-growing, clump-forming, perennial succulent. H ¾in (1.5cm), S 6in (15cm). Has pea-shaped, dark spotted, blue-green leaves, each with a sunken fissure at the tip. Produces cream flowers, ¾in (1.5cm) across, in fall. Z10–11 (min. 41°F/5°C)

CONSOLIDA

Larkspur

RANUNCULACEAE

Genus of annuals, providing excellent cut flowers. Needs sun and fertile, well-drained soil. Support stems of tall-growing plants with sticks. Propagate by seed sown outdoors in spring, or in early fall in mild areas. Protect young plants from slugs and snails. ☉ Seeds are poisonous if ingested. **C. ajacis**, syn. *C. ambigua*, *Delphinium consolida*. Fast-growing, upright, branching annual. Giant forms, H to 4ft (1.2m), S 1ft (30cm); dwarf, H and S 1ft (30cm). All have feathery, mid-green leaves and, throughout summer, spikes of rounded, spurred flowers. Z2–11. **Dwarf Hyacinth Series** cultivars have spikes of tubular flowers in shades of pink, mauve, blue or white. **Giant Imperial Series** (Giant larkspur) bear spikes of rounded, spurred, double flowers in pink, blue or white. **C. ambigua**. See *C. ajacis*.

CONVALLARIA

Lily-of-the-valley

ASPAGACEAE

Genus of spring-flowering, rhizomatous perennials. Prefers partial shade and humus-rich, moist soil. Propagate by division after flowering or in fall. ☉ The seeds of *C. majalis* may cause mild stomach upset if ingested. **C. majalis** illus. p.255. **‘Flore Pleno’** is a low-growing, rhizomatous perennial. H 9–12in (23–30cm), S indefinite. Sprays of small, very fragrant, pendent, bell-shaped flowers that are double and white open in spring. Ovate to lance-shaped leaves are mid- to dark green. **‘Fortin’s Giant’** see ‘Géant de Fortin’. **‘Géant de Fortin’** (syn. *C.m.* ‘Fortin’s Giant’), H 18in (45cm), has larger flowers and leaves that emerge a little earlier. **var. rosea**, H 8in (20cm), has pale mauve-pink flowers. Z2–9

CONVOLVULUS

CONVOLVULACEAE

Genus of upright, climbing or scrambling annuals and perennials, and evergreen

shrubs or subshrubs. Grow in sun and in poor to fertile, well-drained soil. Deadhead to prolong flowering. Propagate by seed sown outdoors in midspring for hardy plants, or under glass in spring for tender plants, perennials and subshrubs by softwood cuttings in late spring or summer. **C. althaeoides** illus. p.362. **C. cneorum** illus. p.149. **C. mauritanicus**. See *C. sabatius*. **C. minor**. See *C. tricolor*. **C. purpureus**. See *Ipomoea purpurea*. **C. sabatius**, syn. *C. mauritanicus*, illus. p.342. **C. tricolor**, syn. *C. minor*. Moderately fast-growing, upright, bushy or climbing annual. H 8–12in (20–30cm), S 8in (20cm). Has ovate to lance-shaped, mid-green leaves. In summer, bears saucer-shaped, blue or white flowers, 1in (25cm) across, with yellowish-white throats. Tall, climbing forms, H to 10ft (3m), have flowers to 4in (10cm) across. Z9–11. **‘Blue Flash’** (bushy) illus. p.314. **‘Royal Ensign’**, H to 12in (30cm), produces deep blue flowers, to 2in (5cm) across; grow in a hanging basket.

COPIAPOA

CACTACEAE

Genus of slow-growing, perennial cacti with funnel-shaped, yellow flowers. Many species have large tap roots. Needs partial shade and very well-drained soil. Propagate by seed or grafting in spring or summer. **C. cinerea** illus. p.489. **C. coquimbana**. Clump-forming, spherical, then columnar, perennial cactus. H to 1ft (30cm), S 3ft (1m). Dark gray-green stem has 10–17 ribs. Areoles each bear 8–10 dark brown radial spines and 1 or 2 sturdier central spines. Yellow flowers, 1¼in (3cm) across, are borne in summer. Slow to form clumps. Z10–11 (min. 41°F/5°C) **C. echinoides**. Flattened spherical, perennial cactus, ribbed like a sea urchin. H 6in (15cm), S 4in (10cm). Solitary gray-green stem bears dark brown spines, 1¼in (3cm) long, which soon fade to gray. In summer, produces pale yellow flowers, 1½in (4cm) across. Z10–11 (min. 41°F/5°C) **C. marginata**. Clump-forming, perennial cactus. H 2ft (60cm), S 1ft (30cm). Gray-green stem bears very close-set areoles with dark-tipped, pale brown spines, to 1¼in (3cm) long. Has yellow flowers, ¾–2in (2–5cm) across, in spring–summer. Z10–11 (min. 41°F/5°C)

COPROSMA

RUBIACEAE

Genus of evergreen shrubs and trees, grown for their foliage and fruits. Separate male and female plants are needed to obtain fruits. Prefers full light and well-drained soil. Water containerized specimens freely in summer, moderately at other times. Propagate by seed in spring or by semiripe cuttings in late summer. **C. baueri** of gardens. See *C. repens*. **C. baueriana**. See *C. repens*. **C. x kirkii ‘Variegata’** illus. p.152. **C. repens**, syn. *C. baueri* of gardens, *C. baueriana*, illus. p.457. **‘Picturata’** is an evergreen, spreading, then erect shrub.

H and S to 6ft (2m). Has broadly, leathery, lustrous, rich green leaves with a central cream to yellow blotch. In late spring, bears insignificant flowers, followed, on female plants, by egg-shaped, orange fruits from late summer to fall. Z8–11

CORDYLINE

ASPAGACEAE

Genus of evergreen shrubs or treelike, woody-stemmed perennials, grown primarily for their foliage, although some also have decorative flowers. Provide fertile, well-drained soil and full light or partial shade. Water potted plants moderately, less in winter. Propagate by seed or suckers in spring or by stem cuttings in summer. Red spider mite may be a nuisance. **C. australis**, syn. *Dracaena australis* (New Zealand cabbage palm). Slow-growing, evergreen, sparsely branched tree. H 50ft (15m) or more, S 15ft (5m) or more. Each stem is crowned by a rosette of strap-shaped leaves, 1–3ft (30–100cm) long. Bears small, scented, white flowers in large, open panicles in summer and, in fall, globose, white fruits. Z9–11. **‘Atropurpurea’** illus. p.451. **‘Torbay Dazzler’** has leaves with bold, cream stripes and margins. **‘Veitchii’** has long, sword-shaped leaves with a red base and midrib. **C. fruticosa**, syn. *C. terminalis* (Good-luck plant, Ti tree). Slow-growing, evergreen, upright shrub, sparingly branched and suckering. H 6–12ft (2–4m), S 3–6ft (1–2m). Broadly lance-shaped, glossy, deep green leaves are 1–2ft (30–60cm) long. Produces branched panicles of small, white, purplish or reddish flowers in summer. Z11–12. (min. 55°F/13°C). **‘Baptistii’** has deep green foliage with pink and yellow stripes and spots. **C. indivisa**, syn. *Dracaena indivisa*. Slow-growing, evergreen, erect tree or shrub. H 10ft (3m) or more, S 6ft (2m). Bears lance-shaped, green leaves, 2–6ft (0.6–2m) long, orange-brown veined above, blue-gray tinted beneath. In summer, tiny, star-shaped, white flowers in dense clusters, 2ft (60cm) or more long, are followed by tiny, spherical, blue-purple fruits. Z9–11 **C. terminalis**. See *C. fruticosa*.

COREOPSIS

Tickseed

ASTERACEAE/COMPOSITAE

Genus of annuals and perennials, grown for their daisylike flower heads. Needs full sun and fertile, well-drained soil. Propagate annuals by seed in spring; *C. lanceolata* by seed or division in spring; *C. auriculata* ‘Superba’, *C. lanceolata* ‘Goldfink’ and *C. grandiflora* ‘Badengold’ by softwood cuttings or division in spring or summer; and *C. verticillata* by division in spring. **C. auriculata ‘Superba’**. Bushy perennial. H and S 18in (45cm). Daisylike, rich yellow flower heads, with central, purple blotches, are borne in summer. Ovate to lance-shaped leaves are lobed and light green. Some plants grown as *C. auriculata* are the closely related annual *C. basalis*. Z4–8 **C. ‘Goldfink’**. See *C. lanceolata* ‘Goldfink’.